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HDOC Day-22 — Pattern Programs in C

1. Algorithm

Step 1: Start the program.

Step 2: Display the menu:

1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit

Step 3: Read the user's choice.

Step 4: Use `switch(choice)`.

Step 5: According to the selected option:

- `choice == 1` → Call `solidSquare()`
- `choice == 2` → Call `solidRectangle()`
- `choice == 3` → Call `rowNumber()`
- `choice == 4` → Call `columnNumber()`
- `choice == 5` → Exit
- Otherwise → Display "Invalid choice!"

Step 6: Continue displaying the menu until the user selects Exit.

Step 7: Stop.

2. Test Case Table

Test Case	Choice	Input	Expected Result	Actual Result	Status
1	1	n = 4	4 × 4 solid square	4 × 4 solid square	Pass
2	2	m = 3, n = 5	3 × 5 solid rectangle	3 × 5 solid rectangle	Pass
3	3	m = 4, n = 4	Rows contain 1s, 2s, 3s, 4s	Same as expected	Pass
4	4	m = 4, n = 4	Each row contains 1 2 3 4	Same as expected	Pass
5	6	—	Invalid choice!	Invalid choice!	Pass
6	5	—	Program terminates	Program terminates	Pass

3. Complete C Program

```
#include <stdio.h>

void solidSquare()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

void solidRectangle()
{
    int m, n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &m);

    printf("Enter number of columns: ");
    scanf("%d", &n);

    for(i = 1; i <= m; i++)
    {
        for(j = 1; j <= n; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}
```

```

void rowNumber()
{
    int m, n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &m);

    printf("Enter number of columns: ");
    scanf("%d", &n);

    for(i = 1; i <= m; i++)
    {
        for(j = 1; j <= n; j++)
        {
            printf("%d ", i);
        }
        printf("\n");
    }
}

void columnNumber()
{
    int m, n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &m);

    printf("Enter number of columns: ");
    scanf("%d", &n);

    for(i = 1; i <= m; i++)
    {
        for(j = 1; j <= n; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

int main()
{
    int choice;

    do
    {
        printf("\n--- PATTERN MENU ---\n");
        printf("1. Solid Square Pattern\n");
        printf("2. Solid Rectangular Pattern\n");
        printf("3. Row-Number Pattern\n");
        printf("4. Column-Number Pattern\n");
        printf("5. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
    }
}

```

```

switch(choice)
{
    case 1:
        solidSquare();
        break;

    case 2:
        solidRectangle();
        break;

    case 3:
        rowNumber();
        break;

    case 4:
        columnNumber();
        break;

    case 5:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice! Please enter 1 to 5.\n");
}

} while(choice != 5);

return 0;
}

```

4. Compilation

Compile the C program

```
cc day22.c
```

If the program contains no syntax errors, the compilation completes successfully.

To execute the program:

```
./a.out
```

5. Execution

Execution 1 — Solid Square Pattern

```

(kali㉿kali)-[~/Desktop]
$ cc day22.c

(kali㉿kali)-[~/Desktop]
$ ./a.out

— PATTERN MENU —
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
5. Exit
Enter your choice: 1
Enter number of rows: 4
* * * *
* * * *
* * * *
* * * *

```

Execution 2 — Solid Rectangular Pattern

```

Enter your choice: 2
Enter number of rows: 3
Enter number of columns: 5
* * * * *
* * * * *
* * * * *

```

Execution 3 — Row-Number Pattern

```

Enter your choice: 3
Enter number of rows: 4
Enter number of columns: 4
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

```

Execution 4 — Column-Number Pattern

```

Enter your choice: 4
Enter number of rows: 4
Enter number of columns: 4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

```

Execution 5 — Invalid Choice

```

5. Exit
Enter your choice: 6
Invalid choice! Please enter 1 to 5.

```

Execution 6 — Exit

```

5. Exit
Enter your choice: 5
Exiting program...

```

6. Result

The C program was **successfully compiled and executed**. All four patterns were generated correctly using **nested loops, switch-case, and parameterless functions** as required. All prepared test cases passed successfully.

Overall Result: PASS